EXP NO-6
DATE-

ERROR DETECTION AND CORRECTION USING HAMMING CODE

AIM: To write a program to implement error detection and correction using the Hamming code.

ALGORITHM:

1) Read the binary data bits.
2) Count the number of data bits m
3) Find the number of parity bits r using $2^r \geq m+r+1$
4) Create positions for Hamming code.
5) Place parity bits at position 1, 2, 4, 8, ...
6) Calculate each parity bit using XOR.
7) Display the Hamming code.
8) Read the wired Hamming code.
9) Check the parity bits again.
10) The position of the parity bits that fail to find the error positions.
11) If the error position is not zero, flip the bit to correct the error.
12) Display the Hamming code.

CODE:

```
import java.util.Scanner;
public class Hamming code {
static int Findparity Bits (int m)
        int r=0;
while (Math.pow(2,r) < m+r+1){
    r+1;
}
return r;

static string encode (String data){
    int m = data.length();
    int r = find parity Bits(m);
    int n = m+r;

int[] code = new int[n+1];
        int j=0;
for (int i=1, i<n; i++){
if ((i or(i-1)) != 0){
  code[i]=data.charAt(j)-'0';
    j++; } }
for (int p=0; p<r; p++){
  int pos= (int) Math.pow(2,p);
      int parity=0;
for (int i=1; i<=n;i++){
    if ((i & pos) != 0){
  parity = parity ^ code[i];
  } } code Epos] = parity;
  }
  String result = " ";
for ( int i=1; i<=n; i++){
   result + = code[i];}
return result; }
```

```
static String Correct Error (String received)
 { int n = received.length();
   int [] code = new int[n+1];
for (int i=1; i<=n; i++){
   code[i] = received.charAt(i-1)-'0';
    } int r=0;
while (Math.pow(2,r) < n+1){
        r++; } int errposition=0;
for (int p=0; p<r; p++){
   int pos = (int) Math.pow(2,p);
    int parity=0;
for (int i=1; i<=r; i++){
    if ((i & position) != 0) {
       parity = parity ^ code[i];
      } }
     if (parity != 0) {
        errorpost = position; } }
 if (errorpos ==0){
system.out.println("No error detected");
}else { system.out.println(" Error
                detected at pos." +
                    errorpos);
  if (code[errorpos] ==0) {
     code[errorpos] = 1; }
else { code[errorpos] = 0; }
System.out.println("Error corrected");
} string corrected = " ";
  for (int i=1; i<=n; i++ ) {
    corrected + = code[i]; }
     return corrected; }
```

```
public static void main (String[] args)
{ Scanner sc = new Scanner
                (System.in);
  System.out.println("Enter data bit:");
    String data = sc.nextLine();
  String hammingcode = encode(data);
  System.out.println ("Generated
    Hamming code: " + hammingcode);
  System.out.println("Enter received
    code:");
  String received = sc.nextLine();
  String corrected = correctError
                (received);
  System.out.println ("Corrected
    Hamming code: " +, corrected);
    sc.close(); } }
```

OUTPUT:

```
Enter data bit : 1011
Generated Hamming code : 0110011
Enter received code : 0110111
Error detected at position : 5
Error corrected.
Corrected Hamming code: 0110011
```

RESULT: The Hamming code program was successfully implemented in Java.